

Monoidal su-categories

Matt Wilson

Université Paris-Saclay, CNRS, ENS Paris-Saclay, Inria, CentraleSupélec,
Laboratoire Méthodes Formelles, France

Giulio Chiribella

HKU-Oxford Joint Laboratory for Quantum Information and Computation

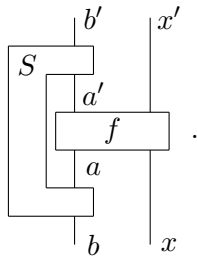
Abstract

We introduce monoidal su-categories, an abstract categorical notion of single-input higher-order process over a monoidal category. The definition separates a base category $\mathcal{C}$ of lower-order processes from a monoidal category $\mathcal{V}$ of holes or supermaps and axiomatizes the compatibility needed for partial application to bipartite processes. For a fixed monoidal base $\mathcal{C}$, monoidal su-categories, monoidal su-functors, and monoidal su-natural transformations form a 2-category $\mathbf{MonSuCat}_{\mathcal{C}}$. We then show that the category $\mathbf{Optic}[\mathcal{C}]$ of coend optics is 2-initial in this 2-category, giving a universal-property characterisation of coend optics as the minimal monoidal theory of single-hole contexts.

1 Introduction

Higher-order processes arise naturally across classical and quantum theoretical computer science. The framework of higher-order quantum operations provides a common language for quantum aspects of the agent-environment interaction in terms of quantum networks, quantum causal correlations, quantum processes with memory, and quantum games [6, 5, 2, 33, 35, 19]. In applied category theory, closely related hole constructions [36, 34, 8, 7, 31] support bidirectional transformations [34, 16, 1], along with compositional approaches to classical game theory, learning, and cybernetics [17, 20, 15, 4].

Many concrete constructions of holes are already available, including coend optics [36, 21], combs [5, 21], higher-order causal processes [29, 38, 39, 27, 26], and strong-natural transformations [22, 45, 44]. A unifying feature among these constructions is local application, more commonly referred to in the quantum-information literature as completeness. At the level of states, local application appears as complete positivity: an operation must remain valid when tensored with the identity on an arbitrary ancilla [32]. At higher order, the same principle underlies quantum superchannels [6, 5], where a legitimate transformation must be applicable to part of any multipartite process, as in the following intuitive picture



Despite its salience, this completeness principle for higher-order transformations does not yet have a clear isolated categorical formulation.

To move towards a stable categorical algebra of holes, we isolate and formulate completeness, that is, local application, in categorical terms. To this end, we introduce monoidal su-categories

(where “su” refers to supermaps) as an abstract categorical model of single-input higher-order processes on a monoidal category. Our focus is on the categorical algebra needed to talk about local application, coherence, and comparison between different hole constructions.

Theorem 1.1. *For every fixed monoidal category $\mathcal{C}$, monoidal su-categories over $\mathcal{C}$ form a 2-category $\mathbf{MonSuCat}_{\mathcal{C}}$.*

We also give a direct graphical interpretation of the axioms and use it to establish a structural theorem on categories of holes. More precisely, we prove the following.

Theorem 1.2. *The monoidal su-category of coend optics $\mathbf{Optic}[\mathcal{C}]$ over a symmetric monoidal category $\mathcal{C}$ is a 2-initial object in $\mathbf{MonSuCat}_{\mathcal{C}}$.*

This work sits between two existing lines of approach. On one side, a number of higher-order constructions land in variants of $*$ -autonomous and BV-like settings [29, 39, 38, 22, 23, 45, 27, 26, 2, 9, 3, 25], where multi-input holes and local application can be studied simultaneously. On the other side, recent work attempts to isolate the *essential* structure of such constructions for them to be interpreted as returning legitimate theories of higher-order processes, using enrichment [43] together with polycategorical and Frobenius-like cotensors [42, 41], though at present this essential structure is treated only in the strict setting. Monoidal su-categories isolate the single-hole local-application fragment common to these settings. This further restriction allows us to tractably relax strictness, introducing coherences and defining functors and natural transformations for comparing and characterising theories of holes, a crucial step towards a full categorical algebra for holes which incorporates multiple inputs.

2 Monoidal categories

We recall briefly the usual abstract algebraic model for processes with both sequential and parallel composition; standard references are [30, 28, 11]. Throughout the paper, $\mathcal{C}$ denotes a base category of lower-order processes and $\mathcal{V}$ denotes a category of higher-order maps. We reserve boldface for named constructions such as $\mathbf{Optic}[\mathcal{C}]$, $\mathbf{Comb}[\mathcal{C}]$, $\mathbf{Slot}[\mathcal{C}]$, and $\mathbf{MonSuCat}_{\mathcal{C}}$.

To specify a category $\mathcal{C}$ one gives a collection of objects $a, b, \dots$ together with a collection of morphisms $f : a \rightarrow b$ for each pair of objects a, b . The set of morphisms of type $a \rightarrow b$ is typically denoted $\mathcal{C}(a, b)$, and for each a, b, c there is a sequential composition rule $\circ_{abc} : \mathcal{C}(a, b) \times \mathcal{C}(b, c) \rightarrow \mathcal{C}(a, c)$ which is associative in the sense that $(h \circ g) \circ f = h \circ (g \circ f)$. Note that subscripts on $\circ_{abc}$ are dropped whenever the context is clear. Sequential composition is also unital, meaning that there exists an identity morphism $1_a : a \rightarrow a$ for each object a satisfying $1_b \circ f = f = f \circ 1_a$.

A strict monoidal category is a category equipped with a parallel composition on objects which lifts to morphisms. More precisely, each pair of objects a, b defines a new object $a \otimes b$, and each pair of morphisms $f : a \rightarrow a'$ and $g : b \rightarrow b'$ defines a new morphism $f \otimes g : a \otimes b \rightarrow a' \otimes b'$. The tensor is associative so that $(a \otimes b) \otimes c = a \otimes (b \otimes c)$ and $(f \otimes g) \otimes h = f \otimes (g \otimes h)$, and it has a unit object I such that $I \otimes a = a = a \otimes I$. Finally, sequential and monoidal composition interact through the interchange law

$$(f \otimes g) \circ (f' \otimes g') = (f \circ f') \otimes (g \circ g').$$

Most naturally occurring monoidal categories are not strict. In general monoidal categories associativity is relaxed up to a natural isomorphism $\alpha_{abc} : (a \otimes b) \otimes c \cong a \otimes (b \otimes c)$, and unitality is relaxed to the existence of natural isomorphisms $\lambda_a : I \otimes a \cong a$ and $\rho_a : a \otimes I \cong a$. These natural isomorphisms are required to satisfy coherence conditions referred to as the triangle and pentagon laws.

Monoidal categories find application in quantum information because they provide the abstract algebra behind circuit-based models of quantum computation and quantum processes

[11, 24]. Circuit diagrams are sound and complete for monoidal categories [28, 37], meaning that any equality derivable from purely topological moves on a circuit diagram is true by the defining equations of a monoidal category and vice versa. Circuit diagrams of this kind in category theory are typically referred to as string diagrams [28, 37].

In string-diagrammatic notation, objects are drawn as wires and morphisms as boxes between those wires:

$$\begin{array}{c} \boxed{f} \\ \hline \end{array}.$$

Sequential and monoidal composition are denoted respectively by

$$\begin{array}{c} \boxed{g} \\ \boxed{f} \end{array} \quad \text{and} \quad \begin{array}{cc} \boxed{f} & \boxed{g} \end{array}.$$

Finally, the unit object I is left unlabelled, being absorbed graphically as empty space.

3 Monoidal su-categories

We now introduce the single-hole structure studied in the rest of the paper. Intuitively, $\mathcal{C}$ contains ordinary processes, $\mathcal{V}$ contains higher-order maps between holes, the object $[a, a'] \in \mathcal{V}$ is the abstract hole with input a and output a' , the isomorphism θ identifies ordinary processes with states of holes, and the structural map $\mu^{\boxtimes}$ expresses the possibility of applying a higher-order map locally to one tensor factor of a bipartite process.

Definition 3.1 (Monoidal su-category). A monoidal su-category consists of:

- A pair of monoidal categories $(\mathcal{V}, \mathcal{C})$
- A functor $[-, =] : \mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathcal{V}$
- A natural isomorphism $\theta : \mathcal{C}(c, d) \cong \mathcal{V}(I_{\mathcal{V}}, [c, d])$
- A natural isomorphism $\mu_{aa'bb'}^{\boxtimes} : [a \otimes b, a' \otimes b'] \rightarrow [a, a'] \boxtimes [b, b']$

such that $\theta(1_{I_{\mathcal{C}}}) : I_{\mathcal{V}} \rightarrow [I_{\mathcal{C}}, I_{\mathcal{C}}]$ is an isomorphism and the following diagrams commute:

- splitting property:

$$\begin{array}{ccc} I_{\mathcal{V}} & \xrightarrow{\lambda_{I_{\mathcal{V}}}^{-1}} & I_{\mathcal{V}} \boxtimes I_{\mathcal{V}} \\ \theta(1_{a \otimes b}) \downarrow & & \downarrow \theta(1_a) \boxtimes \theta(1_b) \\ [a \otimes b, a \otimes b] & \xrightarrow{\mu_{(a,b),(a,b)}^{\boxtimes}} & [a, a] \boxtimes [b, b] \end{array}$$

- mutual coherence between lower and higher associators

$$\begin{array}{ccc} & [c \otimes (d \otimes e), (c' \otimes d') \otimes e'] & \\ \swarrow [\alpha_{cde}, 1] & & \searrow [1, \alpha_{c'd'e'}] \\ [(c \otimes d) \otimes e, (c' \otimes d') \otimes e'] & & [c \otimes (d \otimes e), c' \otimes (d' \otimes e')] \\ \mu_{(c \otimes d)(c' \otimes d')ee'} \downarrow & & \mu_{cc'(d \otimes e)(d' \otimes e')} \downarrow \\ [c \otimes d, c' \otimes d'] \boxtimes [e, e'] & & [c, c'] \boxtimes [d \otimes e, d' \otimes e'] \\ \mu_{cc'dd'} \boxtimes 1 \downarrow & & 1 \boxtimes \mu_{dd'ee'} \downarrow \\ ([c, c'] \boxtimes [d, d']) \boxtimes [e, e'] & \xrightarrow{\alpha_{[c,c'] [d,d'] [e,e']}} & [c, c'] \boxtimes ([d, d'] \boxtimes [e, e']) \end{array}$$

- mutual coherence between lower and higher unitors:

$$\begin{array}{ccc}
[I_{\mathcal{C}} \otimes c, I_{\mathcal{C}} \otimes d] & \xleftarrow{[\lambda_c, 1_{I_{\mathcal{C}} \otimes d}]} & [c, I_{\mathcal{C}} \otimes d] \\
\downarrow \mu_{(I_{\mathcal{C}}, c), (I_{\mathcal{C}}, d)}^{\boxtimes} & & \downarrow [1_c, \lambda_d] \\
[I_{\mathcal{C}}, I_{\mathcal{C}}] \boxtimes [c, d] & & [c, d] \\
\searrow \theta(1_{I_{\mathcal{C}}})^{-1} \boxtimes 1 & & \nearrow \lambda_{[c, d]} \\
& I_{\mathcal{V}} \boxtimes [c, d] &
\end{array}$$

Let us now unpack and directly interpret each feature of the definition of monoidal-su categories. In line with the enrichment-based approach of [43], which models multi-hole structure, a monoidal su-category separates the lower-order process theory $\mathcal{C}$ from the category $\mathcal{V}$ of holes or supermaps. The functor $[-, =]$ sends each pair of objects a, a' in $\mathcal{C}$ to the hole object $[a, a']$ in $\mathcal{V}$:

$$\begin{array}{c} | \\ [a, a'] \end{array} \sim \begin{array}{c} | a' \\ | a \end{array}.$$

Morphisms of type $S : [a, a'] \rightarrow [b, b']$ then represent maps between holes, or supermaps, and through the isomorphism θ every state of type $I_{\mathcal{V}} \rightarrow [a, a']$ corresponds to an ordinary morphism $a \rightarrow a'$ in $\mathcal{C}$:

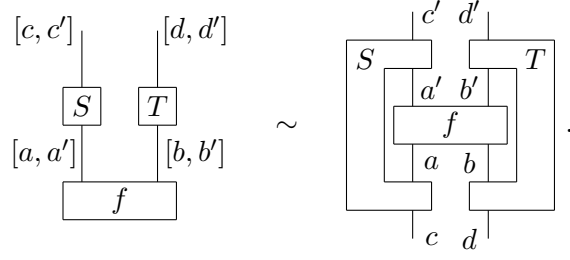
$$\begin{array}{c} [b, b'] \\ \downarrow \\ \boxed{S} \\ \downarrow \\ [a, a'] \end{array} \sim \begin{array}{c} b' \\ \downarrow \\ \boxed{S} \\ \downarrow \\ a' \\ \downarrow \\ a \\ \downarrow \\ b \end{array}. \quad \begin{array}{c} [a, a'] \\ \downarrow \\ \boxed{f} \end{array} \sim \begin{array}{c} a' \\ \downarrow \\ \boxed{f} \\ \downarrow \\ a \end{array} \sim \begin{array}{c} a' \\ \downarrow \\ \boxed{f} \\ \downarrow \\ a \end{array}.$$

Note that composition in $\mathcal{V}$ represents nesting of supermaps between holes

$$\begin{array}{c} [c, c'] \\ \downarrow \\ \boxed{T} \\ \downarrow \\ [b, b'] \\ \downarrow \\ \boxed{S} \\ \downarrow \\ [a, a'] \end{array} \sim \begin{array}{c} c' \\ \downarrow \\ \boxed{T} \\ \downarrow \\ b' \\ \downarrow \\ \boxed{S} \\ \downarrow \\ a' \\ \downarrow \\ a \\ \downarrow \\ b \\ \downarrow \\ c \end{array}.$$

The isomorphism $\mu^{\boxtimes}$ is the key feature for modelling completeness: it means that states of type $I_{\mathcal{V}} \rightarrow [a, a'] \boxtimes [b, b']$ precisely correspond to bipartite morphisms in $\mathcal{C}$, that is, to morphisms of type $a \otimes b \rightarrow a' \otimes b'$. This gives categorical semantics to partial application on arbitrary

bipartite morphisms:



The remaining conditions in the definition of a monoidal su-category are coherence conditions, so they do not admit much direct diagrammatic interpretation beyond identifying different ways of reading empty space and formal bracketing.

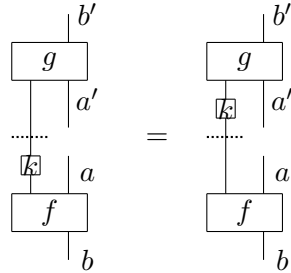
Example 3.2 (Optics). Following [36, 21], the symmetric monoidal category **Optic** $[\mathcal{C}]$ has object pairs (a, a') of objects of $\mathcal{C}$ and hom-sets

$$\mathbf{Optic}[\mathcal{C}]((a, a'), (b, b')) := \int^{m \in \mathcal{C}} \mathcal{C}(b, m \otimes a) \times \mathcal{C}(m \otimes a', b').$$

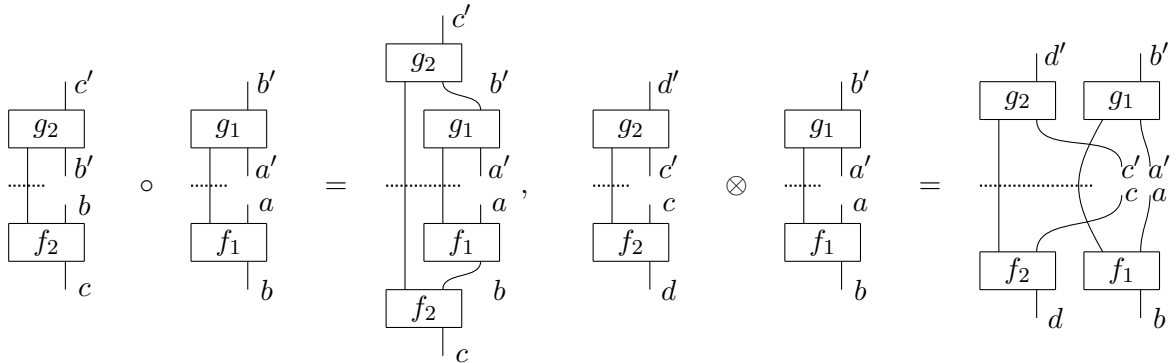
Concretely, a morphism is represented by an auxiliary object m together with a pair of maps

$$(f : b \longrightarrow m \otimes a, g : m \otimes a' \longrightarrow b'),$$

quotiented by equivalence under sliding, that is, equivalence under $((1 \otimes u)f, g) \cong (f, g(1 \otimes u))$. Graphically, coend optics can be denoted by



With parallel composition on objects defined by $(c, c') \otimes (d, d') := (c \otimes d, c' \otimes d')$ and with sequential and parallel composition of morphisms defined by



The symmetric monoidal category **Optic** $[\mathcal{C}]$ can be equipped with the structure of a symmetric monoidal su-category as follows.

- The functor $[-, =]_{\cong} : \mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathbf{Optic}[\mathcal{C}]$ is the identity on objects and sends a pair of morphisms $(f : b \rightarrow a, g : a' \rightarrow b')$ to the corresponding memoryless optic, with auxiliary object $I_{\mathcal{C}}$, induced by f and g through the unitors of $\mathcal{C}$.

- The natural isomorphism $\mathbf{Optic}[\mathcal{C}]((I_C, I_C), (c, c')) \cong \mathcal{C}(c, c')$, where a state is represented by an auxiliary-free optic and the unitors identify that representative with an ordinary morphism $c \rightarrow c'$.
- Noting that $[a, a'] \boxtimes [b, b'] = (a, a') \boxtimes (b, b') = (a \otimes b, a' \otimes b') = [a \otimes b, a' \otimes b']$, we take $\mu_{aa'bb'}$ to be the identity morphism.

Graphically, these assignments are represented by

$$[f, g] \mapsto (f, g) := \frac{I}{I} \begin{array}{c} \boxed{g} \\ | \\ b' \\ \vdots \\ b \\ \boxed{f} \\ | \\ c \end{array}, \quad f \mapsto \frac{c'}{c} \begin{array}{c} \boxed{f} \\ | \\ c \\ \vdots \\ c \end{array} = \frac{I}{I} \begin{array}{c} | \\ \vdots \\ c' \\ \boxed{f} \\ | \\ c \end{array}.$$

The required coherences are inherited from the coherence isomorphisms in $\mathcal{C}$, and the splitting law reduces to

$$\frac{a}{\vdots} \frac{I}{I} \boxtimes \frac{b}{\vdots} \frac{I}{I} = \frac{a}{\vdots} \frac{b}{\vdots} \frac{I}{I}.$$

Example 3.3 (Quantum superchannels). Traditionally, the definition of quantum superchannels is given directly in terms of the Choi–Jamiołkowski isomorphism [6]. To keep the presentation self-contained, we instead follow [40] and use simple isomorphisms available in finite-dimensional Hilbert spaces, which are in general used to construct the Choi–Jamiołkowski isomorphism. We note that the set of quantum channels $\mathbf{QC}(A, A')$ from Hilbert space A to Hilbert space A' is a subset $\mathbf{QC}(A, A') \subseteq \mathbf{L}(\mathbf{L}(A), \mathbf{L}(A'))$ of the set of linear maps from $\mathbf{L}(A)$ to $\mathbf{L}(A')$, where for any X , $\mathbf{L}(X) := \mathbf{L}(X, X)$. The local application $(S \otimes_u i)(\mathcal{E})$ of any linear map $S: \mathbf{L}(\mathbf{L}(A), \mathbf{L}(A')) \rightarrow \mathbf{L}(\mathbf{L}(B), \mathbf{L}(B'))$ to an element

$$\mathcal{E} \in \mathbf{QC}(A \otimes X, A' \otimes X') \subseteq \mathbf{L}(\mathbf{L}(A \otimes X), \mathbf{L}(A' \otimes X'))$$

can be defined using the following isomorphisms (which are inherited from the compact closure of the category $\mathbf{L}$ of finite-dimensional Hilbert spaces, giving for each A, B a natural isomorphism $u_{AB}: \mathbf{L}(A \otimes B, A \otimes B) \cong \mathbf{L}(A, A) \otimes \mathbf{L}(B, B)$):

$$\begin{array}{ccc} \mathbf{L}(\mathbf{L}(A), \mathbf{L}(A')) \otimes \mathbf{L}(\mathbf{L}(X), \mathbf{L}(X')) & \xrightarrow{S \otimes i} & \mathbf{L}(\mathbf{L}(B), \mathbf{L}(B')) \otimes \mathbf{L}(\mathbf{L}(X), \mathbf{L}(X')) \\ \uparrow u_{LALA' LXLX'}^{-1} & & \uparrow u_{LBLB' LXLX'} \\ \mathbf{L}(\mathbf{L}(A) \otimes \mathbf{L}(X), \mathbf{L}(A') \otimes \mathbf{L}(X')) & & \mathbf{L}(\mathbf{L}(B) \otimes \mathbf{L}(X), \mathbf{L}(B') \otimes \mathbf{L}(X')) \\ \uparrow \mathbf{L}(u_{AX}, u_{A'X'}^{-1}) & & \uparrow \mathbf{L}(u_{BX}^{-1}, u_{B'X'}) \\ \mathbf{L}(\mathbf{L}(A \otimes X), \mathbf{L}(A' \otimes X')) & \xrightarrow{(S \otimes_u i)} & \mathbf{L}(\mathbf{L}(B \otimes X), \mathbf{L}(B' \otimes X')) \end{array} \quad (2.8)$$

A linear map S is a quantum superchannel if, for every $\mathcal{E} \in \mathbf{QC}(A \otimes X, A' \otimes X')$, one has

$$(S \otimes_u i)(\mathcal{E}) \in \mathbf{QC}(B \otimes X, B' \otimes X').$$

In other words, a quantum superchannel is a linear map that *completely* preserves quantum channels: it can be *locally applied* to a channel and is guaranteed to return a new quantum channel.

In the appendix we give several further constructions of theories of higher-order maps that form monoidal su-categories.

3.1 The 2-category of monoidal su-categories

We now define the morphisms and 2-morphisms that compare monoidal su-categories over a fixed lower-order theory $\mathcal{C}$ while allowing the higher-order category $\mathcal{V}$ to vary.

Theorem 3.4 (The 2-category $\mathbf{MonSuCat}_{\mathcal{C}}$). *For every monoidal category $\mathcal{C}$, the following data define a 2-category $\mathbf{MonSuCat}_{\mathcal{C}}$.*

- 0-cells are monoidal su-categories $(\mathcal{V}, \mathcal{C})$.
- 1-cells are monoidal su-functors $(\mathcal{F}, \mu^{\mathcal{F}}) : (\mathcal{V}, \mathcal{C}) \rightarrow (\mathcal{V}', \mathcal{C})$, where $\mathcal{F} : \mathcal{V} \rightarrow \mathcal{V}'$ is a monoidal functor and $\mu_{a,a'}^{\mathcal{F}} : [a, a'] \rightarrow \mathcal{F}[a, a']$ is a natural isomorphism such that, writing $\zeta_{cc'dd'} := (\mu_{cc'dd'}^{\boxtimes})^{-1}$ and $(u^{\mathcal{F}}, \phi^{\mathcal{F}})$ for the monoidal structure of $\mathcal{F}$, the diagrams

$$\begin{array}{ccc} I_{\mathcal{V}'} & \xrightarrow{\theta'(1_{I_{\mathcal{C}}})} & [I_{\mathcal{C}}, I_{\mathcal{C}}] \\ u^{\mathcal{F}} \downarrow & & \downarrow \mu_{I_{\mathcal{C}}, I_{\mathcal{C}}}^{\mathcal{F}} \\ \mathcal{F}(I_{\mathcal{V}}) & \xrightarrow{\mathcal{F}(\theta(1_{I_{\mathcal{C}}}))} & \mathcal{F}[I_{\mathcal{C}}, I_{\mathcal{C}}] \end{array}$$

and

$$\begin{array}{ccc} [c \otimes d, c' \otimes d'] & \xleftarrow{\zeta_{cc'dd'}} & [c, c'] \boxtimes [d, d'] \\ \mu_{c \otimes d, c' \otimes d'}^{\mathcal{F}} \downarrow & & \downarrow \mu_{c, c'}^{\mathcal{F}} \boxtimes \mu_{d, d'}^{\mathcal{F}} \\ \mathcal{F}[c \otimes d, c' \otimes d'] & \xleftarrow{\mathcal{F}(\zeta_{cc'dd'}) \circ \phi_{[c, c'], [d, d']}^{\mathcal{F}}} & \mathcal{F}[c, c'] \boxtimes \mathcal{F}[d, d'] \end{array}$$

commute. Composition is given by

$$\mu^{\mathcal{G}\mathcal{F}} := \mathcal{G}(\mu^{\mathcal{F}}) \circ \mu^{\mathcal{G}}.$$

and

$$u^{\mathcal{G}\mathcal{F}} := \mathcal{G}(u^{\mathcal{F}}) \circ u^{\mathcal{G}}.$$

- 2-cells are monoidal su-natural transformations $\eta : (\mathcal{F}, \mu^{\mathcal{F}}) \Rightarrow (\mathcal{G}, \mu^{\mathcal{G}})$, that is, monoidal natural transformations $\eta : \mathcal{F} \Rightarrow \mathcal{G}$ satisfying

$$\begin{array}{ccc} & [c, c'] & \\ \mu_{c, c'}^{\mathcal{F}} \swarrow & & \searrow \mu_{c, c'}^{\mathcal{G}} \\ \mathcal{F}[c, c'] & \xrightarrow{\eta_{[c, c']}} & \mathcal{G}[c, c'] \end{array}$$

for every $c, c' \in \mathcal{C}$.

Proof. Given in Appendix A.1. □

Morphisms in the fixed-base 2-category $\mathbf{MonSuCat}_{\mathcal{C}}$ give a direct way to compare algebraic theories of holes over the same lower-order process theory $\mathcal{C}$. As we will see, they give a way to frame the idea that coend-optics can be embedded inside *any* theory of concurrent holes.

4 Graphical notation for monoidal su-categories

In this section we introduce a basic string-diagrammatic notation for monoidal su-categories and monoidal su-functors. The notation is intentionally conservative: a supermap of type $[a, a'] \rightarrow [b, b']$ is drawn as an ordinary process in $\mathcal{V}$ rather than as a literal box with a hole. A sound and complete hole-based graphical calculus is left for future work.

Recall that a monoidal su-category consists of a pair of monoidal categories $(\mathcal{V}, \mathcal{C})$. We draw string diagrams in $\mathcal{V}$, but use the functor $[-, =] : \mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathcal{V}$ to label certain wires by hole objects. More precisely, we depict the action of $[-, =]$ on a pair of morphisms (f, g) by

$$[f, g] := \left| \begin{array}{c} \vdots \\ \boxed{f} \quad \boxed{g} \\ \vdots \end{array} \right|.$$

We write the natural isomorphism $\theta : \mathcal{C}(c, d) \cong \mathcal{V}(I_{\mathcal{V}}, [c, d])$ as

$$\theta(f) := \left| \begin{array}{c} \vdots \\ \boxed{f} \\ \vdots \end{array} \right| = \left| \begin{array}{c} \vdots \\ \boxed{f} \\ \vdots \end{array} \right|.$$

We write the structural isomorphism $\mu_{aa'bb'}^{\boxtimes} : [a \otimes b, a' \otimes b'] \rightarrow [a, a'] \boxtimes [b, b']$ as

$$\mu^{-1} := \left| \begin{array}{c} \vdots \\ \text{X} \\ \vdots \end{array} \right|, \quad \mu := \left| \begin{array}{c} \text{X} \\ \vdots \\ \vdots \end{array} \right|,$$

with inverse equations

$$\left| \begin{array}{c} \vdots \\ \text{X} \\ \vdots \end{array} \right| = \left| \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right|, \quad \left| \begin{array}{c} \text{X} \\ \vdots \\ \vdots \end{array} \right| = \left| \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right|,$$

and naturality displayed by

$$\left| \begin{array}{c} \vdots \\ \text{X} \\ \vdots \end{array} \right| = \left| \begin{array}{c} \vdots \\ \boxed{f} \quad \boxed{f'} \\ \vdots \end{array} \right|.$$

The unit state $\theta(1_{I_{\mathcal{C}}}) : I_{\mathcal{V}} \rightarrow [I_{\mathcal{C}}, I_{\mathcal{C}}]$ is also an isomorphism; we depict its inverse as

$$\theta(1_{I_{\mathcal{C}}})^{-1} := \left| \begin{array}{c} \vdots \\ \text{---} \\ \vdots \end{array} \right|$$

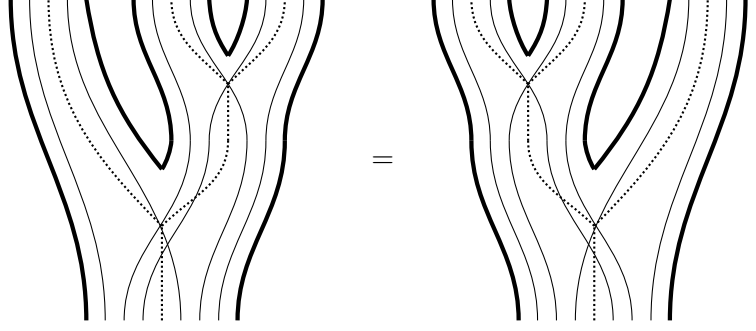
so that

$$\left| \begin{array}{c} \vdots \\ \text{---} \\ \vdots \end{array} \right| = 1.$$

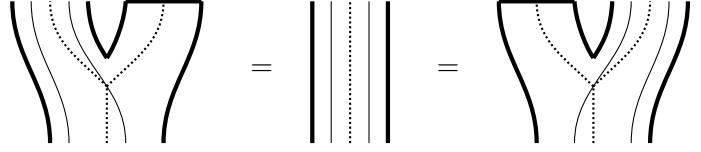
The axioms used repeatedly later are the splitting law

$$\left| \begin{array}{c} \vdots \\ \text{X} \\ \vdots \end{array} \right| = \left| \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right|,$$

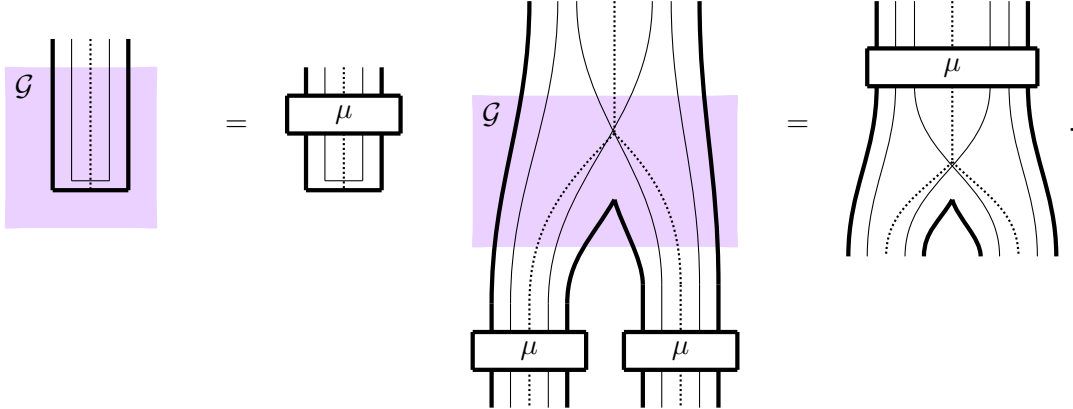
associativity for splitting



and unitality for splitting



A monoidal su-functor can then be expressed graphically using the following functor-box notation, subject to the additional properties



5 A universal property for coend optics

In what follows we fix a symmetric monoidal category $\mathcal{C}$ and work in $\mathbf{MonSuCat}_{\mathcal{C}}$, so every 1-cell acts as the identity on the lower-order category $\mathcal{C}$. We therefore write an object simply as $\mathcal{V}$ when no confusion can arise.

We can now state the second main theorem of the paper, which identifies coend optics as the universal minimal theory of single-hole contexts over $\mathcal{C}$.

Theorem 5.1 (Initiality of optics). *$\mathbf{Optic}[\mathcal{C}]$ is a 2-initial object in $\mathbf{MonSuCat}_{\mathcal{C}}$.*

Proof. Define a functor

$$\mathcal{O}_{\mathcal{V}} : \mathbf{Optic}[\mathcal{C}] \longrightarrow \mathcal{V}$$

on objects by

$$\mathcal{O}_{\mathcal{V}}(a, a') := [a, a'].$$

For a representative (f, g, m) of an optic $(a, a') \rightarrow (b, b')$, with

$$f : b \rightarrow m \otimes a, \quad g : m \otimes a' \rightarrow b',$$

define the corresponding morphism in $\mathcal{V}$ by the graphical composite

$$\mathcal{O}_m(f, g) := \text{[Diagram: A string diagram with two boxes labeled } f \text{ and } g \text{ at the top. } f \text{ has a solid line entering from the left and a solid line exiting to the left. } g \text{ has a solid line entering from the right and a solid line exiting to the right. A dotted line connects the two boxes. Below them, a solid line forms a loop that crosses the dotted line. The diagram is enclosed in a frame with a thick left boundary and a thick right boundary.]}$$

To see that this assignment descends to the coend, first verify the corresponding commutative diagram for the cowedge:

$$\begin{array}{ccc} \mathcal{C}(b, z \otimes a) \times \mathcal{C}(z' \otimes a', b') & \xrightarrow{\mathcal{C}(b, u \otimes 1_a) \times 1} & \mathcal{C}(b, z' \otimes a) \times \mathcal{C}(z' \otimes a', b') \\ \downarrow 1 \times \mathcal{C}(u \otimes 1_{a'}, b') & & \downarrow \mathcal{O}_{(a, a'), (b, b'), z'} \\ \mathcal{C}(b, z \otimes a) \times \mathcal{C}(z \otimes a', b') & \xrightarrow{\mathcal{O}_{(a, a'), (b, b'), z}} & \mathcal{V}([a, a'], [b, b']) \end{array}$$

and hence obtain the unique morphism

$$\int^{z \in \mathcal{C}} \mathcal{C}(b, z \otimes a) \times \mathcal{C}(z \otimes a', b') \longrightarrow \mathcal{V}([a, a'], [b, b']).$$

In string-diagrammatic language, this is precisely the sliding equation

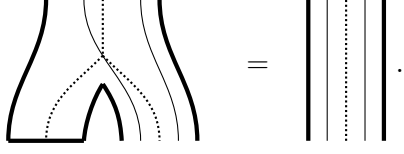
$$\text{[Diagram: Two string diagrams separated by an equals sign. The left diagram has boxes } f, g \text{ and } v \text{ with solid and dotted lines. The right diagram is identical but with the lines rearranged to show a sliding property. Both diagrams are enclosed in a frame with thick boundaries.]}$$

This defines $\mathcal{O}_\gamma$ on hom-sets. We now check that it is functorial. For preservation of the identity, the relevant commutative diagram reads:

$$\begin{array}{ccccc} [c, c'] & \xrightarrow{1_{[c, c']}} & & [c, c'] & \\ & \searrow \lambda_{[c, c']} & & \swarrow \lambda_{[c, c']} & \\ & & I_{\mathcal{V}} \boxtimes [c, c'] & & \\ & & \downarrow \theta(1_{I_{\mathcal{C}}}) \boxtimes 1_{[c, c']} & & \\ & & [I_{\mathcal{C}}, I_{\mathcal{C}}] \boxtimes [c, c'] & & \\ & & \downarrow \mu_{(I_{\mathcal{C}}, c), (I_{\mathcal{C}}, c')}^{\boxtimes} & & \\ & & [I_{\mathcal{C}} \otimes c, I_{\mathcal{C}} \otimes c'] & & \\ & \swarrow [\lambda_c, \lambda_{c'}] & & \searrow [\lambda_c, \lambda_{c'}] & \\ [c, c'] & \xrightarrow{1_{[c, c']}} & & [c, c'] & \end{array}$$

$\mathcal{O}_{\mathcal{V}}(1_{(c, c')})$ on the left, $1_{[c, c']}$ on the right.

Diagrammatically, this simply reads



For preservation of composition, let $\omega : (c, c') \rightarrow (d, d')$ be represented by (f, g, z) and let $\omega' : (d, d') \rightarrow (e, e')$ be represented by (f', g', y) . Their composite coend optic has auxiliary object $y \otimes z$ given concretely by

$$e \xrightarrow{f'} y \otimes d \xrightarrow{1_y \otimes f} y \otimes (z \otimes c) \xrightarrow{\alpha_{yzc}^{-1}} (y \otimes z) \otimes c$$

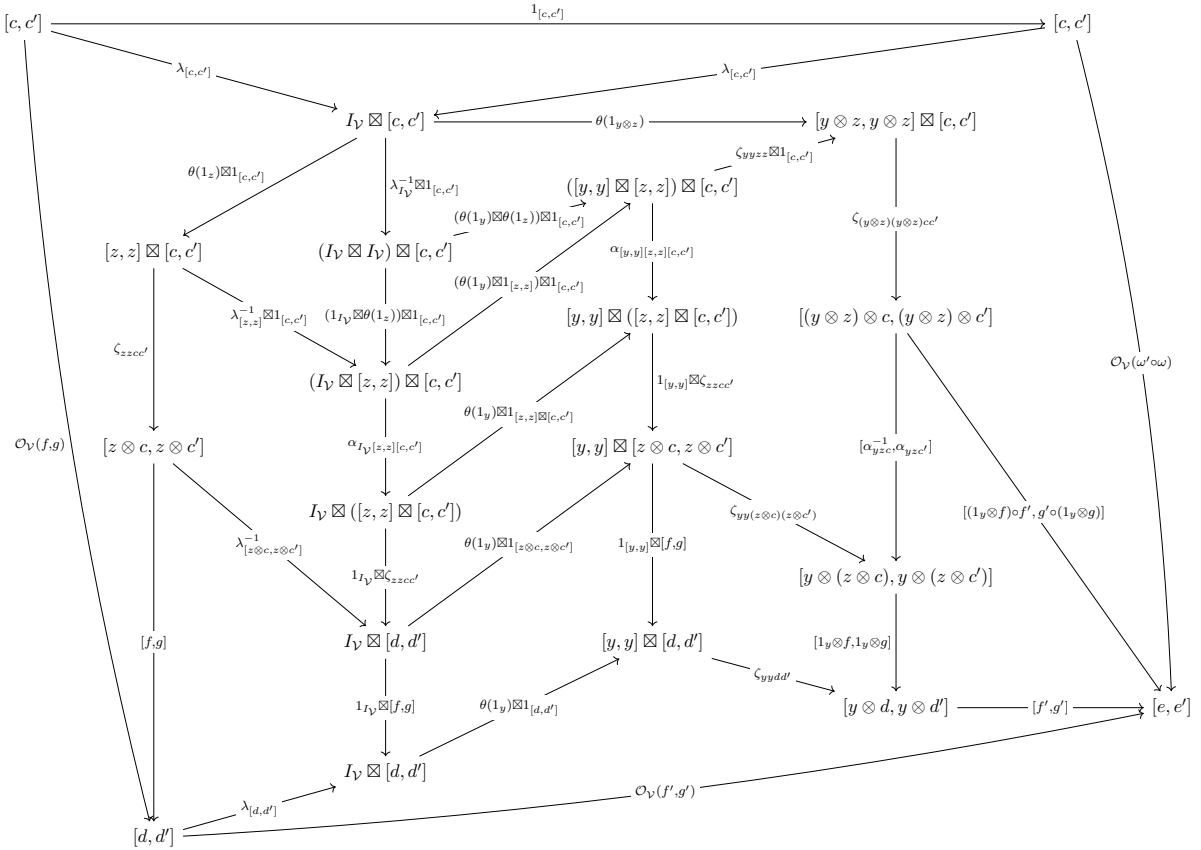
and

$$(y \otimes z) \otimes c' \xrightarrow{\alpha_{yzc'}} y \otimes (z \otimes c') \xrightarrow{1_y \otimes g} y \otimes d' \xrightarrow{g'} e'.$$

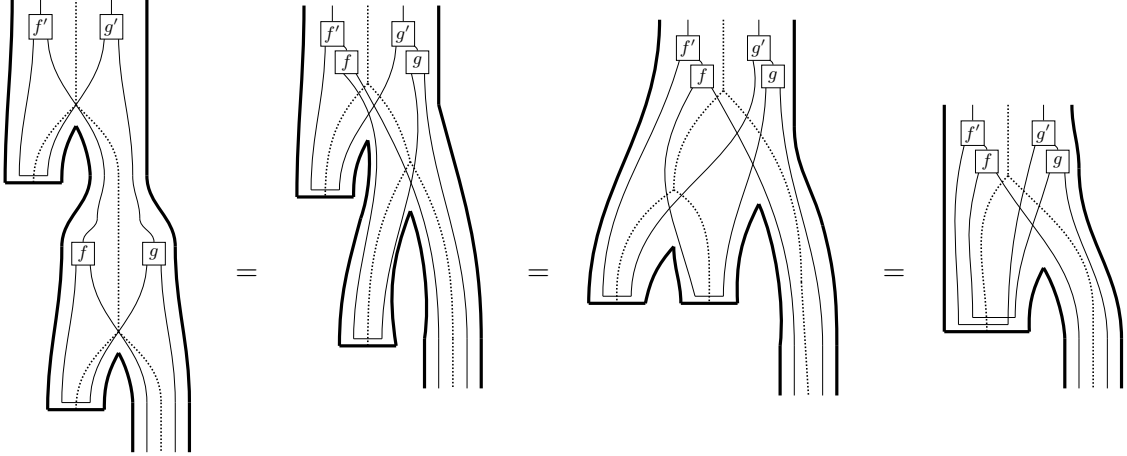
The commutative-diagram verification of

$$\mathcal{O}_V(\omega' \circ \omega) = \mathcal{O}_V(\omega') \circ \mathcal{O}_V(\omega)$$

is:



which is more intuitive diagrammatically:



To equip this with the structure of a monoidal su-functor, take

$$u^{\mathcal{O}} = \theta(1_{I_C}) := \begin{array}{c} \vdots \\ \boxed{} \end{array}$$

along with $\mu_{a,a'}^{\mathcal{O}} := 1_{[a,a']}$ and $\phi_{[c,c'],[d,d']}^{\mathcal{O}} := \zeta_{cc'dd'}$. The required coherence between $\mu^{\mathcal{O}}$ and $u^{\mathcal{O}}$ trivialises to

$$\begin{array}{ccc} I_{\mathcal{V}} & \xrightarrow{\theta(1_{I_C})} & [I_C, I_C] \\ u^{\mathcal{O}} = \theta(1_{I_C}) \downarrow & & \downarrow \mu_{I_C, I_C}^{\mathcal{O}} = 1_{[I_C, I_C]} \\ [I_C, I_C] & \xrightarrow{\mathcal{O}_{\mathcal{V}}(1_{(I_C, I_C)}) = 1_{[I_C, I_C]}} & [I_C, I_C] \end{array}$$

and the tensor coherence trivialises to

$$\begin{array}{ccc} [c \otimes d, c' \otimes d'] & \xleftarrow{\zeta_{cc'dd'}} & [c, c'] \boxtimes [d, d'] \\ \mu_{c \otimes d, c' \otimes d'}^{\mathcal{O}} = 1 \downarrow & & \downarrow \mu_{c, c'}^{\mathcal{O}} \boxtimes \mu_{d, d'}^{\mathcal{O}} = 1 \boxtimes 1 \\ [c \otimes d, c' \otimes d'] & & [c, c'] \boxtimes [d, d'] \\ & \swarrow \mathcal{O}_{\mathcal{V}}(\zeta_{cc'dd'}) = 1_{[c \otimes d, c' \otimes d']} \quad \searrow \phi_{[c, c'], [d, d']}^{\mathcal{O}} = \zeta_{cc'dd'} & \\ & [c \otimes d, c' \otimes d'] & \end{array}$$

so $\mathcal{O}_{\mathcal{V}}$ is indeed a monoidal su-functor.

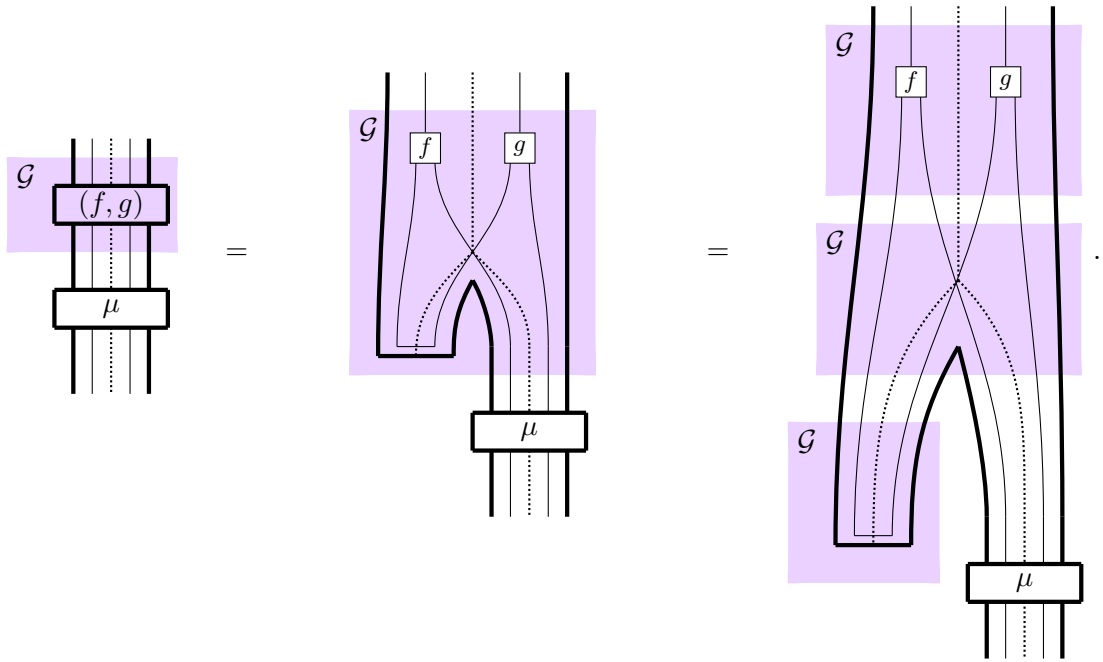
Now let $\mathcal{G} : \mathbf{Optic}[\mathcal{C}] \rightarrow \mathcal{V}$ be any other monoidal su-functor. Since $[a, a'] = \mathcal{O}_{\mathcal{V}}(a, a')$, the natural isomorphisms $\mu_{a,a'}^{\mathcal{G}} : [a, a'] \rightarrow \mathcal{G}(a, a')$ define a comparison 2-cell (which is furthermore an isomorphism)

$$\eta : \mathcal{O}_{\mathcal{V}} \Rightarrow \mathcal{G}, \quad \eta_{(a,a')} := \mu_{a,a'}^{\mathcal{G}}.$$

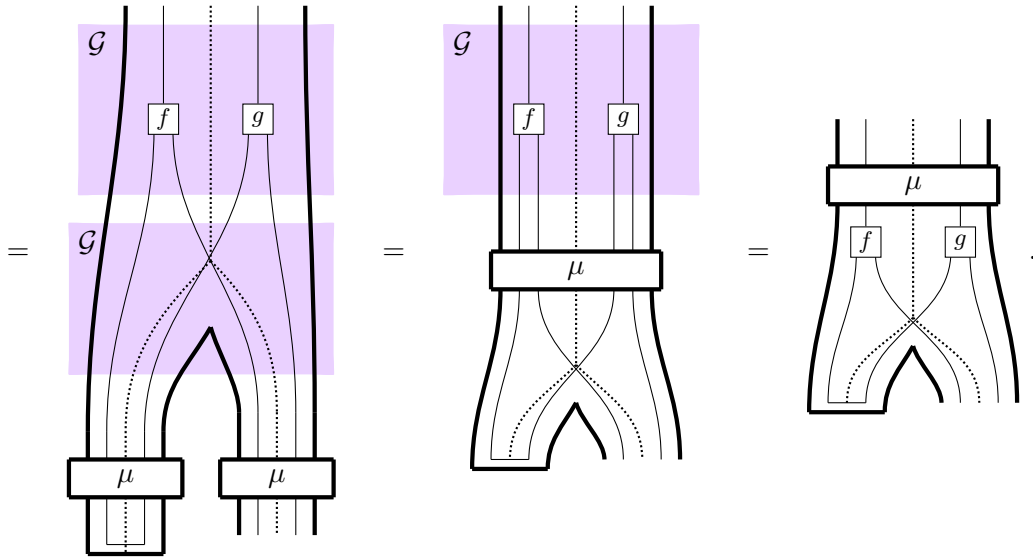
Naturality is the requirement that for every optic (f, g) the diagram

$$\begin{array}{ccc} [c, d] & \xrightarrow{\mu_{c,d}^{\mathcal{G}}} & \mathcal{G}(c, d) \\ \mathcal{O}_{\mathcal{V}}(f, g) \downarrow & & \downarrow \mathcal{G}(f, g) \\ [c', d'] & \xrightarrow{\mu_{c',d'}^{\mathcal{G}}} & \mathcal{G}(c', d') \end{array}$$

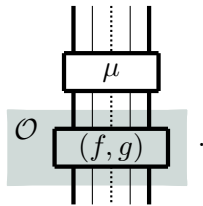
commutes, and this may again be verified with string diagrams:



then, using su-functoriality of $\mathcal{G}$,



which is precisely the lower-left path of



The monoidality of η is the commutativity of

$$\begin{array}{ccc}
[c, c'] \boxtimes [d, d'] & \xrightarrow{\zeta_{cdc'd'}} & [c \otimes d, c' \otimes d'] \\
\downarrow \mu_{cc'}^{\mathcal{G}} \boxtimes \mu_{dd'}^{\mathcal{G}} & \nearrow \mathcal{G}((c, c') \otimes (d, d')) & \downarrow \mu_{c \otimes d, c' \otimes d'}^{\mathcal{G}} \\
\mathcal{G}[c, c'] \boxtimes \mathcal{G}[d, d'] & \xrightarrow[\phi_{[c, c'], [d, d']}^{\mathcal{G}}]{} & \mathcal{G}([c \otimes d, c' \otimes d'])
\end{array}$$

$\nearrow \phi_{[c, c'], [d, d']}^{\mathcal{G}}$
 $\searrow \overline{\mathcal{G}}(1_{(c, c') \otimes (d, d')})$

which commutes by su-functoriality of $\mathcal{G}$. The su-naturality triangle reads directly as

$$\begin{array}{ccc}
& [c, d] & \\
1_{[c, d]} \swarrow & & \searrow \mu_{cd}^{\mathcal{G}} \\
[c, d] & \xrightarrow[\eta_{(c, d)} := \mu_{cd}^{\mathcal{G}}]{} & \mathcal{G}(c, d)
\end{array}$$

For uniqueness, if

$$\tau : \mathcal{O}_{\mathcal{V}} \Rightarrow \mathcal{G}$$

is any monoidal su-natural transformation, then for every (a, a') one has

$$\tau_{(a, a')} \circ \mu_{a, a'}^{\mathcal{O}} = \mu_{a, a'}^{\mathcal{G}}.$$

Since $\mu_{a, a'}^{\mathcal{O}} = 1_{[a, a']}$, it follows that

$$\tau_{(a, a')} = \mu_{a, a'}^{\mathcal{G}} = \eta_{(a, a')},$$

and so η is the unique comparison 2-cell, and **Optic** $[\mathcal{C}]$ is 2-initial in **MonSuCat** $_{\mathcal{C}}$. $\square$

This universal property for optics within theories of supermaps is closely related to the universal property for coend optics arising from the adjunction between (pro)duoidal categories and normal (pro)duoidal categories [14]. For any symmetric monoidal category $\mathcal{C}$, the category $\mathcal{C}^{op} \times \mathcal{C}$ carries a canonical produoidal structure, and its normalisation under that adjunction is the category of coend optics. The result of [14] therefore yields an initiality statement for coend optics in the corresponding comma category of produoidal functors out of $\mathcal{C}^{op} \times \mathcal{C}^1$.

Despite the conceptual similarity, the two setups differ in several important ways, suggesting that there may be a more general framework unifying them. On the produoidal side one obtains a laxator of the form $[a, a'] \boxtimes [b, b'] \rightarrow [a \otimes b, a' \otimes b']$, whereas monoidal su-categories are organised around the inverse splitting isomorphism $\mu^{\boxtimes} : [a \otimes b, a' \otimes b'] \rightarrow [a, a'] \boxtimes [b, b']$. Likewise, morphisms in the produoidal comma category preserve the hom-functor $[-, =]$ on the nose, while monoidal su-functors preserve the corresponding structure only up to coherent isomorphism. Finally, monoidal su-categories include the state/process isomorphism θ , while the produoidal perspective keeps track of an additional *sequencing* tensor product which in practice handles multi-input higher-order transformations. At a high level, monoidal su-categories focus on the completeness of higher-order transformations, whereas produoidal categories focus on their multi-hole aspect. Taken together, these perspectives suggest that in a categorical semantics for multi-hole higher-order transformations with completeness included, one could still reasonably hope to frame coend optics as suitably initial.

¹The authors are grateful to James Hefford for identifying, unpacking, and explaining this similarity.

6 Conclusion and Outlook

Monoidal su-categories provide a categorical algebra of single-hole higher-order processes. The framework isolates the data needed to interpret a hole object $[a, a']$, the state/process correspondence θ , and the splitting map $\mu^\boxtimes$ that makes local application to bipartite processes possible. Over a fixed symmetric monoidal base $\mathcal{C}$, these structures assemble into the 2-category $\mathbf{MonSuCat}_{\mathcal{C}}$.

Several directions now seem especially promising:

- *Coherence for monoidal su-categories.* Monoidal su-categories involve two interacting families of coherence, and we conjecture that every monoidal su-category is equivalent to one in which both the underlying category $\mathcal{C}$ and the higher-order category $\mathcal{V}$ are strictified along with the isomorphisms θ and μ . Stating such a theorem will require extending the notion of monoidal su-functor to allow both the underlying and higher-order category to vary.
- *An expressive graphical language for higher-order quantum operations.* The ZX calculus equips traditional monoidal categories with gadgets (representing a core feature of quantum mechanics, the existence of complementary observables), that allow for expressive and automated graphical reasoning about correctness and compilation in first-order quantum protocols [13, 12]. By establishing a suitable replacement for the traditional circuit model, monoidal su-categories give a natural starting point for building an expressive graphical language for reasoning with higher-order quantum protocols.
- *Dynamical resource theories.* The notion of monoidal subcategory is used to formulate in categorical terms what it means to be a resource theory [10], and so the notion of monoidal su-subcategory suggests a clean categorical approach to dynamical resource theories [18], where the resources to be transformed are themselves processes.
- *Beyond the single-hole setting.* The next natural step is to extend the framework from single-hole to multi-input higher-order structure. The enrichment-based approach of [43], the polycategorical approach of [42], and recent logical/categorical work on duoidal and BV-style structure [9, 38, 23, 25, 14] will inform further development towards a broader non-strict categorical semantics of multi-input higher-order processes.

Most broadly, the definition of monoidal su-categories captures succinctly a simple compositional idea in formal categorical algebra, providing a stable foundation on which more domain-specific tools can be built and through which cross-domain pollination between formal computer science and the foundations of quantum information theory can be facilitated.

7 Acknowledgements

The authors are grateful to James Hefford for useful discussions, particularly regarding the relationship between the universal property for optics established here and the adjunction between produoidal and normal-produoidal categories. Part of MW’s contribution was carried out at UCL and part at CentraleSupélec. While based at University College London, MW was funded by the Engineering and Physical Sciences Research Council [grant number EP/W524335/1].

References

- [1] Faris Abou-Saleh, James Cheney, Jeremy Gibbons, James McKinna, and Perdita Stevens. Introduction to bidirectional transformations. *Lecture Notes in Computer Science*, 9715 LNCS:1–28, 2018. doi:10.1007/978-3-319-79108-1{_}1.

- [2] Alessandro Bisio and Paolo Perinotti. Theoretical framework for higher-order quantum theory. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 475(2225):20180706, 2019. URL: <http://dx.doi.org/10.1098/rspa.2018.0706>, doi:10.1098/rspa.2018.0706.
- [3] Richard Blute, Prakash Panangaden, and Sergey Slavnov. Deep inference and probabilistic coherence spaces. *Applied Categorical Structures*, 20:209–228, 2012. doi:10.1007/s10485-010-9241-0.
- [4] Michele Capucci, Bruno Gavranović, Jules Hedges, and Eigil Fjeldgren Rischel. Towards foundations of categorical cybernetics. *Electronic Proceedings in Theoretical Computer Science*, 372:235–248, 2022. doi:10.4204/EPTCS.372.17.
- [5] Giulio Chiribella, Giacomo M. D’Ariano, and Paolo Perinotti. Theoretical framework for quantum networks. *Physical Review A - Atomic, Molecular, and Optical Physics*, 80(2), 4 2009. doi:10.1103/PhysRevA.80.022339.
- [6] Giulio Chiribella, Giacomo Mauro D’Ariano, and Paolo Perinotti. Transforming quantum operations: Quantum supermaps. *EPL (Europhysics Letters)*, 83(3):30004, 2008. doi:10.1209/0295-5075/83/30004.
- [7] Bryce Clarke. A diagrammatic approach to symmetric lenses. *Electronic Proceedings in Theoretical Computer Science*, 333:79–91, 2 2021. doi:10.4204/EPTCS.333.6.
- [8] Bryce Clarke, Derek Elkins, Jeremy Gibbons, Fosco Loregian, Bartosz Milewski, Emily Pillmore, and Mario Román. Profunctor optics, a categorical update, 2020. arXiv:2001.07488. URL: <https://arxiv.org/abs/2001.07488>, arXiv:2001.07488.
- [9] J.R.B. Cockett and R.A.G. Seely. Weakly distributive categories. *Journal of Pure and Applied Algebra*, 114(2):133–173, 1997. URL: <https://www.sciencedirect.com/science/article/pii/0022404995001603>, doi:10.1016/0022-4049(95)00160-3.
- [10] Bob Coecke, Tobias Fritz, and Robert W. Spekkens. A mathematical theory of resources. *Inf. Comput.*, 250:59–86, 2016. doi:10.1016/j.ic.2016.02.008.
- [11] Bob Coecke and Aleks Kissinger. *Picturing Quantum Processes: A First Course in Quantum Theory and Diagrammatic Reasoning*. Cambridge University Press, 2017. doi:10.1017/9781316219317.
- [12] Bob Coecke and Aleks Kissinger. *Picturing quantum processes: A first course in quantum theory and diagrammatic reasoning*. Cambridge University Press, 3 2017. doi:10.1017/9781316219317.
- [13] Ross Duncan, Aleks Kissinger, Simon Perdrix, and John van de Wetering. Graph-theoretic Simplification of Quantum Circuits with the ZX-calculus. *Quantum*, 4:279, 2020. doi:10.22331/q-2020-06-04-279.
- [14] Matt Earnshaw, James Hefford, and Mario Román. The produoidal algebra of process decomposition, 2023. URL: <https://arxiv.org/abs/2301.11867>, arXiv:2301.11867.
- [15] Brendan Fong, David I. Spivak, and Rémy Tuyéras. Backprop as functor: A compositional perspective on supervised learning. In *Proceedings of the 34th Annual ACM/IEEE Symposium on Logic in Computer Science, LICS ’19*, pages 1–13, Vancouver, BC, Canada, 2019. IEEE. URL: <https://lics.siglog.org/archive/2019/Tuyras-BackpropasFunctorAc.html>, doi:10.1109/LICS.2019.8785665.

- [16] J. Nathan Foster, Michael B. Greenwald, Jonathan T. Moore, Benjamin C. Pierce, and Alan Schmitt. Combinators for bidirectional tree transformations: A linguistic approach to the view-update problem. *ACM Transactions on Programming Languages and Systems*, 29(3):17, 5 2007. URL: <https://dl.acm.org/doi/10.1145/1232420.1232424>, doi:10.1145/1232420.1232424.
- [17] Neil Ghani, Jules Hedges, Viktor Winschel, and Philipp Zahn. Compositional game theory. *Proceedings of the 33rd Annual ACM/IEEE Symposium on Logic in Computer Science (LICS 2018)*, pages 472–481, 2018. doi:10.1145/3209108.3209165.
- [18] Gilad Gour and Carlo Maria Scandolo. Dynamical Resources, 2021. URL: <https://arxiv.org/abs/2101.01552>, arXiv:2101.01552.
- [19] Gus Gutoski and John Watrous. Toward a general theory of quantum games. In *Proceedings of the thirty-ninth annual ACM symposium on Theory of computing*, STOC07, pages 565–574. ACM, June 2007. URL: <http://dx.doi.org/10.1145/1250790.1250873>, doi:10.1145/1250790.1250873.
- [20] Jules Hedges. Coherence for lenses and open games, 4 2017. URL: <https://arxiv.org/abs/1704.02230>, arXiv:1704.02230.
- [21] James Hefford and Cole Comfort. Coend Optics for Quantum Combs. In *Proceedings ACT 2022*, volume 380, pages 63–76, 2023. doi:10.4204/EPTCS.380.4.
- [22] James Hefford and Matt Wilson. A Profunctorial Semantics for Quantum Supermaps. In *Proceedings of the 39th Annual ACM/IEEE Symposium on Logic in Computer Science, LICS '24*, New York, NY, USA, 2024. Association for Computing Machinery. doi:10.1145/3661814.3662123.
- [23] James Hefford and Matt Wilson. A bv-category of spacetime interventions, 2025. URL: <https://arxiv.org/abs/2502.19022>, arXiv:2502.19022.
- [24] Chris Heunen and Jamie Vicary. Introduction to Categorical Quantum Mechanics. Technical report, University of Oxford, 2013. URL: <https://www.cs.ox.ac.uk/people/jamie.vicary/IntroductionToCategoricalQuantumMechanics.pdf>.
- [25] Kengo Hirata and Takeshi Tsukada. Causality in Pure Quantum Computation with Quantum Control. In Claudia Faggian and Joost-Pieter Katoen, editors, *41st Annual Symposium on Logic in Computer Science (LICS 2026)*, volume 380 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 57:1–57:28, Dagstuhl, Germany, 2026. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. URL: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.LICS.2026.57>, doi:10.4230/LIPIcs.LICS.2026.57.
- [26] Timothée Hoffreumon and Ognian Oreshkov. Projective characterization of higher-order quantum transformations. *Quantum*, 10:1978, January 2026. URL: <http://dx.doi.org/10.22331/q-2026-01-21-1978>, doi:10.22331/q-2026-01-21-1978.
- [27] Anna Jenčová. On the structure of higher order quantum maps, 2024. URL: <https://arxiv.org/abs/2411.09256>, arXiv:2411.09256.
- [28] André Joyal and Ross Street. The geometry of tensor calculus, I. *Advances in Mathematics*, 88(1):55–112, 1991. doi:10.1016/0001-8708(91)90003-P.
- [29] Aleks Kissinger and Sander Uijlen. A categorical semantics for causal structure. *Logical Methods in Computer Science*, 15(3), 2019. doi:10.23638/LMCS-15(3:15)2019.

- [30] Saunders Mac Lane. *Categories for the Working Mathematician*. Graduate Texts in Mathematics. Springer Science+Business Media, New York, NY, 2 edition, September 1998. eBook package: Springer Book Archive; Copyright Springer Science+Business Media New York 1978. doi:10.1007/978-1-4757-4721-8.
- [31] J. López-González and Juan M. Serrano. Towards optic-based algebraic theories: The case of lenses. In *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, volume 11457 LNCS, pages 74–93. Springer Verlag, 6 2019. URL: https://link.springer.com/chapter/10.1007/978-3-030-18506-0_4, doi:10.1007/978-3-030-18506-0_{_}4.
- [32] Michael A. Nielsen and Isaac L. Chuang. *Quantum computation and quantum information*. Cambridge University Press, 2010.
- [33] Ognjan Oreshkov, Fabio Costa, and Časlav Brukner. Quantum correlations with no causal order. *Nature Communications*, 3(1092), 2012. doi:10.1038/ncomms2076.
- [34] Matthew T. Pickering, Jeremy Gibbons, and Nicolas Wu. Profunctor optics: Modular data accessors. *The Art, Science, and Engineering of Programming*, 1(2):7, 2017. URL: <https://programming-journal.org/2017/1/7/>, doi:10.22152/programming-journal.org/2017/1/7.
- [35] Felix A. Pollock, César Rodríguez-Rosario, Thomas Frauenheim, Mauro Paternostro, and Kavan Modi. Non-Markovian quantum processes: complete framework and efficient characterisation. *Physical Review A*, 97(1):012127, Jan 2018. doi:10.1103/PhysRevA.97.012127.
- [36] Mitchell Riley. Categories of Optics, 9 2018. arXiv:1809.00738, doi:10.48550/arXiv.1809.00738.
- [37] Peter Selinger. A survey of graphical languages for monoidal categories. *Lecture Notes in Physics*, 813:289–355, 8 2009. arxiv:0908.3347. URL: https://doi.org/10.1007/978-3-642-12821-9_4, doi:10.1007/978-3-642-12821-9_{_}4.
- [38] Will Simmons and Aleks Kissinger. Higher-Order Causal Theories Are Models of BV-Logic. In Stefan Szeider, Robert Galian, and Alexandra Silva, editors, *47th International Symposium on Mathematical Foundations of Computer Science (MFCS 2022)*, volume 241 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 80:1–80:14, Dagstuhl, Germany, 2022. Schloss Dagstuhl – Leibniz-Zentrum für Informatik. doi:10.4230/LIPIcs.MFCS.2022.80.
- [39] Will Simmons and Aleks Kissinger. A complete logic for causal consistency, 2024. arXiv:2403.09297.
- [40] M. Wilson. *Compositional Frameworks for Supermaps and Causality*. Phd thesis, University of Oxford, 2023.
- [41] Matt Wilson. Higher-order circuits, 2026. URL: <https://arxiv.org/abs/2602.18701>, arXiv:2602.18701.
- [42] Matt Wilson and Giulio Chiribella. Free polycategories for unitary supermaps of arbitrary dimension, 2022. URL: <https://arxiv.org/abs/2207.09180>, arXiv:2207.09180.
- [43] Matt Wilson and Giulio Chiribella. A mathematical framework for transformations of physical processes, 2023. doi:10.48550/arXiv.2204.04319.

- [44] Matt Wilson, Giulio Chiribella, and Aleks Kissinger. Quantum supermaps are characterized by locality, 2022. doi:10.48550/ARXIV.2205.09844.
- [45] Matt Wilson, James Hefford, and Timothée Hoffreumon. Supermaps on generalised theories, 2026. URL: <https://arxiv.org/abs/2602.23865>, arXiv:2602.23865.

A Detailed Proofs and Commutative Diagrams

The main text uses string diagrams where they are more readable. Here we record the full proofs in terms of commutative diagrams where necessary for completeness.

A.1 The 2-category $\mathbf{MonSuCat}_{\mathcal{C}}$

We now give the full proof of Theorem 3.4. Monoidal categories, together with monoidal functors and monoidal natural transformations, form the 2-category $\mathbf{MonCat}$. Adding su-structure, one must check that the additional μ -morphisms are closed under composition and that the corresponding coherence is preserved by units, vertical composition, and whiskering.

For closure under composition, suppose

$$(\mathcal{F}, \mu^{\mathcal{F}}) : (\mathcal{V}, \mathcal{C}) \rightarrow (\mathcal{V}', \mathcal{C}) \quad \text{and} \quad (\mathcal{G}, \mu^{\mathcal{G}}) : (\mathcal{V}', \mathcal{C}) \rightarrow (\mathcal{V}'', \mathcal{C})$$

are monoidal su-functors. We now check the following two diagrams:

$$\begin{array}{ccc} I_{\mathcal{V}''} & \xrightarrow{\theta''(1_{I_{\mathcal{C}}})} & [I_{\mathcal{C}}, I_{\mathcal{C}}] \\ \downarrow u^{\mathcal{G}} & \searrow u^{\mathcal{G}} & \downarrow \mu_{I_{\mathcal{C}}, I_{\mathcal{C}}}^{\mathcal{G}} \\ \mathcal{G}(I_{\mathcal{V}'}) & \xrightarrow{\mathcal{G}(\theta'(1_{I_{\mathcal{C}}}))} & \mathcal{G}[I_{\mathcal{C}}, I_{\mathcal{C}}] \\ \downarrow u^{\mathcal{F}} & \swarrow \mathcal{G}(u^{\mathcal{F}}) & \downarrow \mathcal{G}(\mu_{I_{\mathcal{C}}, I_{\mathcal{C}}}^{\mathcal{F}}) \\ \mathcal{GF}(I_{\mathcal{V}}) & \xrightarrow{\mathcal{GF}(\theta(1_{I_{\mathcal{C}}}))} & \mathcal{GF}[I_{\mathcal{C}}, I_{\mathcal{C}}] \end{array},$$

and

$$\begin{array}{ccccc} [c \otimes d, c' \otimes d'] & \xleftarrow{\zeta_{cc'dd'}} & [c, c'] \boxtimes [d, d'] \\ \downarrow \mu_{c \otimes d, c' \otimes d'}^{\mathcal{G}} & & \downarrow \mu_{c, c'}^{\mathcal{G}} \boxtimes \mu_{d, d'}^{\mathcal{G}} \\ \mathcal{G}[c \otimes d, c' \otimes d'] & & \mathcal{G}[c, c'] \boxtimes \mathcal{G}[d, d'] \\ \downarrow \mu_{c \otimes d, c' \otimes d'}^{\mathcal{GF}} & \swarrow \mathcal{G}(\mu_{c \otimes d, c' \otimes d'}^{\mathcal{F}}) & \downarrow \mathcal{G}(\mu_{c, c'}^{\mathcal{F}}) \boxtimes \mathcal{G}(\mu_{d, d'}^{\mathcal{F}}) \\ \mathcal{GF}[c \otimes d, c' \otimes d'] & \swarrow \mathcal{G}(\zeta_{cc'dd'}) & \mathcal{GF}[c, c'] \boxtimes \mathcal{GF}[d, d'] \\ & \downarrow \mathcal{G}(\mu_{c, c'}^{\mathcal{F}} \boxtimes \mu_{d, d'}^{\mathcal{F}}) & \downarrow \mathcal{G}(\mu_{c, c'}^{\mathcal{F}}) \boxtimes \mathcal{G}(\mu_{d, d'}^{\mathcal{F}}) \\ & \mathcal{G}([c, c'] \boxtimes [d, d']) & \downarrow \mathcal{G}(\mu_{c, c'}^{\mathcal{F}}) \boxtimes \mathcal{G}(\mu_{d, d'}^{\mathcal{F}}) \\ & \downarrow \mathcal{G}(\mu_{c, c'}^{\mathcal{F}} \boxtimes \mu_{d, d'}^{\mathcal{F}}) & \downarrow \mathcal{G}(\mu_{c, c'}^{\mathcal{F}}) \boxtimes \mathcal{G}(\mu_{d, d'}^{\mathcal{F}}) \\ & \mathcal{G}(\mathcal{F}[c, c'] \boxtimes \mathcal{F}[d, d']) & \downarrow \mathcal{G}(\mu_{c, c'}^{\mathcal{F}}) \boxtimes \mathcal{G}(\mu_{d, d'}^{\mathcal{F}}) \\ & \downarrow \mathcal{G}(\phi_{[c, c'], [d, d']}^{\mathcal{F}}) & \downarrow \mathcal{G}(\mu_{c, c'}^{\mathcal{F}}) \boxtimes \mathcal{G}(\mu_{d, d'}^{\mathcal{F}}) \\ & \mathcal{GF}([c, c'] \boxtimes [d, d']) & \downarrow \mathcal{G}(\mu_{c, c'}^{\mathcal{F}}) \boxtimes \mathcal{G}(\mu_{d, d'}^{\mathcal{F}}) \\ & & \downarrow \mathcal{G}(\mu_{c, c'}^{\mathcal{F}}) \boxtimes \mathcal{G}(\mu_{d, d'}^{\mathcal{F}}) \\ & & \mathcal{GF}([c, c'] \boxtimes [d, d']) \end{array}$$

For associativity of composition, note that

$$\mu^{(\mathcal{H}\mathcal{G})^{\mathcal{F}}} = \mathcal{H}\mathcal{G}(\mu^{\mathcal{F}}) \circ \mu^{\mathcal{H}\mathcal{G}} = \mathcal{H}\mathcal{G}(\mu^{\mathcal{F}}) \circ \mathcal{H}(\mu^{\mathcal{G}}) \circ \mu^{\mathcal{H}} = \mathcal{H}(\mu^{\mathcal{G}\mathcal{F}}) \circ \mu^{\mathcal{H}} = \mu^{\mathcal{H}(\mathcal{G}\mathcal{F})}.$$

For the unit 1-cell, one simply takes the identity functor equipped with the identity natural transformation $\mu^{\mathcal{I}} : [-, =] \Rightarrow [-, =]$, with the extra monoidal su-functor coherences satisfied immediately. For unitality of sequential composition, it is immediate that $\mu^{\mathcal{G}\mathcal{I}} = \mu^{\mathcal{G}}$ and $\mu^{\mathcal{I}\mathcal{F}} = \mu^{\mathcal{F}}$.

The identity natural transformation satisfies the required su-coherence by commutativity of

$$\begin{array}{ccc} & [c, c'] & \\ \mu_{c, c'}^{\mathcal{F}} \swarrow & & \searrow \mu_{c, c'}^{\mathcal{F}} \\ \mathcal{F}[c, c'] & \xrightarrow{1_{[c, c']}} & \mathcal{F}[c, c'] \end{array}.$$

Finally, one checks that the extra coherences required for monoidal su-natural transformations are preserved under sequential composition and whiskering. For sequential composition $\tau \circ \eta$ of natural transformations $\eta : \mathcal{F} \Rightarrow \mathcal{G}$ and $\tau : \mathcal{G} \Rightarrow \mathcal{H}$ we have commutativity of

$$\begin{array}{ccccc} & [c, c'] & & & \\ & \mu_{c, c'}^{\mathcal{F}} \swarrow & & \searrow \mu_{c, c'}^{\mathcal{H}} & \\ \mathcal{F}[c, c'] & \xrightarrow{\eta_{[c, c']}} & \mathcal{G}[c, c'] & \xrightarrow{\tau_{[c, c']}} & \mathcal{H}[c, c'] \end{array},$$

whereas for whiskering $\eta \star \tau$ with $\eta : \mathcal{F} \Rightarrow \mathcal{F}'$ and $\tau : \mathcal{G} \Rightarrow \mathcal{G}'$ one may verify coherence for the composite by observing commutativity of

$$\begin{array}{ccccc} & [c, c'] & & & \\ & \downarrow \mu_{c, c'}^{\mathcal{G}} & & & \\ & \mathcal{G}[c, c'] & & & \\ \mu_{c, c'}^{\mathcal{G}\mathcal{F}} \swarrow & & \searrow \tau_{[c, c']} & & \mu_{c, c'}^{\mathcal{G}'\mathcal{F}'} \\ \mathcal{G}(\mu_{c, c'}^{\mathcal{F}}) \swarrow & \mathcal{G}(\mu_{c, c'}^{\mathcal{F}'}) \downarrow & & \searrow \mathcal{G}'(\mu_{c, c'}^{\mathcal{F}'}) & \\ \mathcal{G}\mathcal{F}[c, c'] & \mathcal{G}\mathcal{F}'[c, c'] & & \mathcal{G}'\mathcal{F}'[c, c'] & \\ \mathcal{G}(\eta_{[c, c']}) \swarrow & & \searrow \tau_{\mathcal{F}'[c, c']} & & \\ \mathcal{G}\mathcal{F}[c, c'] & & \mathcal{G}'\mathcal{F}'[c, c'] & & \end{array}$$

($\tau \star \eta$)_[c, c']

B Examples of Monoidal Su-Categories

Example B.1 (Combs). Following [5, 21], the symmetric monoidal category **Comb**[$\mathcal{C}$] has the same objects as **Optic**[$\mathcal{C}$], namely pairs (a, a') , and morphisms

$$\mathbf{Comb}[\mathcal{C}]((a, a'), (b, b')) := \mathbf{Optic}[\mathcal{C}]((a, a'), (b, b')) / \cong_{\text{comb}},$$

where two representatives (f_1, g_1) and (f_2, g_2) are considered equivalent when, for every pair of objects $x, x' \in \mathcal{C}$ and every morphism $\phi : a \otimes x \rightarrow a' \otimes x'$, the induced composites

$$b \otimes x \xrightarrow{f_1 \otimes 1_x} (m_1 \otimes a) \otimes x \cong m_1 \otimes (a \otimes x) \xrightarrow{1_m \otimes \phi} m_1 \otimes (a' \otimes x') \cong (m_1 \otimes a') \otimes x' \xrightarrow{g_1 \otimes 1_{x'}} b' \otimes x'$$

and the corresponding composite built from (f_2, g_2) agree. We can express this requirement graphically by

$$\begin{array}{c} \begin{array}{|c|} \hline g_1 \\ \hline \end{array} \begin{array}{|c|} \hline b' \\ \hline \end{array} \begin{array}{|c|} \hline a' \\ \hline \end{array} \begin{array}{|c|} \hline a \\ \hline \end{array} \begin{array}{|c|} \hline \dots \\ \hline \end{array} \begin{array}{|c|} \hline f_1 \\ \hline \end{array} \begin{array}{|c|} \hline b \\ \hline \end{array} \end{array} \cong_{\text{Comb}} \begin{array}{c} \begin{array}{|c|} \hline g_2 \\ \hline \end{array} \begin{array}{|c|} \hline c' \\ \hline \end{array} \begin{array}{|c|} \hline b' \\ \hline \end{array} \begin{array}{|c|} \hline b \\ \hline \end{array} \begin{array}{|c|} \hline \dots \\ \hline \end{array} \begin{array}{|c|} \hline f_2 \\ \hline \end{array} \begin{array}{|c|} \hline c \\ \hline \end{array} \end{array} \iff \forall x, x', \phi : a \otimes x \rightarrow a' \otimes x' : \begin{array}{c} \begin{array}{|c|} \hline g_1 \\ \hline \end{array} \begin{array}{|c|} \hline b' \\ \hline \end{array} \begin{array}{|c|} \hline a' \\ \hline \end{array} \begin{array}{|c|} \hline a \\ \hline \end{array} \begin{array}{|c|} \hline \phi \\ \hline \end{array} \begin{array}{|c|} \hline f_1 \\ \hline \end{array} \begin{array}{|c|} \hline b \\ \hline \end{array} \begin{array}{|c|} \hline x \\ \hline \end{array} \end{array} = \begin{array}{c} \begin{array}{|c|} \hline g_2 \\ \hline \end{array} \begin{array}{|c|} \hline b' \\ \hline \end{array} \begin{array}{|c|} \hline a' \\ \hline \end{array} \begin{array}{|c|} \hline a \\ \hline \end{array} \begin{array}{|c|} \hline \phi \\ \hline \end{array} \begin{array}{|c|} \hline f_2 \\ \hline \end{array} \begin{array}{|c|} \hline b \\ \hline \end{array} \begin{array}{|c|} \hline x \\ \hline \end{array} \end{array} .$$

Sequential and parallel composition are inherited directly from the corresponding composition rules in the category of optics.

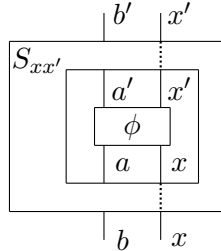
For our next example we refer to a more abstract approach to defining supermaps, namely as locally applicable transformations [44, 45], which are in fact morphisms between strong profunctors [22]. We say that a locally applicable transformation of type

$$S : (a, a') \longrightarrow (b, b')$$

is given by families of functions

$$S_{x, x'} : \mathcal{C}(a \otimes x, a' \otimes x') \longrightarrow \mathcal{C}(b \otimes x, b' \otimes x')$$

indexed by pairs $x, x' \in \mathcal{C}$ and natural over the category of coend-optics, meaning commuting with combs on the environment. Graphically, we may denote a locally applicable transformation by



with naturality condition expressed by

$$\forall x, x', \phi, f, g : a \otimes x \rightarrow a' \otimes x' : \begin{array}{c} \begin{array}{|c|} \hline b' \\ \hline \end{array} \begin{array}{|c|} \hline y' \\ \hline \end{array} \begin{array}{|c|} \hline g \\ \hline \end{array} \begin{array}{|c|} \hline x' \\ \hline \end{array} \begin{array}{|c|} \hline S_{xx'} \\ \hline \end{array} \begin{array}{|c|} \hline a' \\ \hline \end{array} \begin{array}{|c|} \hline x' \\ \hline \end{array} \begin{array}{|c|} \hline \phi \\ \hline \end{array} \begin{array}{|c|} \hline a \\ \hline \end{array} \begin{array}{|c|} \hline x \\ \hline \end{array} \begin{array}{|c|} \hline f \\ \hline \end{array} \begin{array}{|c|} \hline b \\ \hline \end{array} \begin{array}{|c|} \hline y \\ \hline \end{array} \end{array} = \begin{array}{c} \begin{array}{|c|} \hline b' \\ \hline \end{array} \begin{array}{|c|} \hline y' \\ \hline \end{array} \begin{array}{|c|} \hline g \\ \hline \end{array} \begin{array}{|c|} \hline y' \\ \hline \end{array} \begin{array}{|c|} \hline S_{yy'} \\ \hline \end{array} \begin{array}{|c|} \hline a' \\ \hline \end{array} \begin{array}{|c|} \hline x' \\ \hline \end{array} \begin{array}{|c|} \hline \phi \\ \hline \end{array} \begin{array}{|c|} \hline a \\ \hline \end{array} \begin{array}{|c|} \hline x \\ \hline \end{array} \begin{array}{|c|} \hline f \\ \hline \end{array} \begin{array}{|c|} \hline b \\ \hline \end{array} \begin{array}{|c|} \hline y \\ \hline \end{array} \end{array} .$$

Example B.2 (Slots). Locally applicable transformations do not in general form monoidal categories, because the interchange law can fail [42]. Slots are defined as locally applicable transformations satisfying a stronger locality condition, namely commutativity with any locally applicable transformation acting on the environment. Graphically, this condition can be

expressed by

$$\begin{array}{c}
 \begin{array}{|c|c|c|}
 \hline
 a'_2 & b'_2 & x' \\
 \hline
 \begin{array}{|c|c|c|}
 \hline
 a'_1 & b'_2 & x' \\
 \hline
 \begin{array}{|c|c|c|}
 \hline
 a'_1 & b'_1 & x' \\
 \hline
 \begin{array}{|c|c|c|}
 \hline
 \phi \\
 \hline
 a_1 & b_1 & x \\
 \hline
 \end{array} \\
 \hline
 a_1 & b_2 & x \\
 \hline
 \end{array} \\
 \hline
 a_2 & b_2 & x \\
 \hline
 \end{array}
 \end{array}
 \end{array}
 =
 \begin{array}{c}
 \begin{array}{|c|c|c|}
 \hline
 a'_2 & b'_2 & x' \\
 \hline
 \begin{array}{|c|c|c|}
 \hline
 a'_2 & b'_1 & x' \\
 \hline
 \begin{array}{|c|c|c|}
 \hline
 S_{b_1 b'_1 x x'} \\
 \hline
 \begin{array}{|c|c|c|}
 \hline
 a'_1 & b'_1 & x' \\
 \hline
 \begin{array}{|c|c|c|}
 \hline
 \phi \\
 \hline
 a_1 & b_1 & x \\
 \hline
 \end{array} \\
 \hline
 a_2 & b_1 & x \\
 \hline
 \end{array} \\
 \hline
 \end{array} \\
 \hline
 a_2 & b_2 & x \\
 \hline
 \end{array}
 \end{array}
 \end{array}
 .$$

The symmetric monoidal category $\mathbf{Slot}[\mathcal{C}]$ can be equipped with the structure of a symmetric monoidal su-category as follows.

- The functor $[-, =]_{\cong} : \mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathbf{Slot}[\mathcal{C}]$ is defined on objects by $[a, a'] = (a, a')$ and sends a pair $(f : b \rightarrow a, g : a' \rightarrow b')$ to the locally applicable transformation

$$[f, g]_{x, x'}(\phi) := (g \otimes 1_{x'}) \circ \phi \circ (f \otimes 1_x).$$

- The natural isomorphism $\mathbf{Slot}[\mathcal{C}]((I_{\mathcal{C}}, I_{\mathcal{C}}), (c, c')) \cong \mathcal{C}(c, c')$ arises by a Yoneda-like argument. Concretely, note that any locally applicable transformation of type $(I, I) \rightarrow (c, c')$ has the form $S_{xx'}(\phi) = S_{II}(1_I) \otimes \phi$ for some fixed morphism $S_{II}(1_I)$. This gives a way to construct a representative morphism for each locally applicable transformation. Conversely, given any morphism $f : c \rightarrow c'$, one may construct an associated natural transformation $S_{xx'}(\phi) := f \otimes \phi$. It is routine to check that these two constructions are mutually inverse to each other and natural in c, c' .
- The definition of the monoidal product in the category of slots is defined to be $(c, c') \otimes (d, d') := (c \otimes d, c' \otimes d')$.

The coherence axioms for monoidal su-categories are then checked directly.